SIMATS ENGINEERING

ASSESSMENT TOOL 2 – LABORATORY PERFORMANCE QUESTIONS

COURSE CODE: CSA0904 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO4: *Applet Programming*-Applet Architecture - Simple Applet Display Methods Event handling in Java - Delegation Event Model - Event Classes - Sources of Events - Event Listener Interfaces -AWT Component and GUI Design : AWT Classes, Controls, Layout Managers, and Menus.*(BL3,BL4)*
Assessment Tool Weightage: 20%
QUESTIONS: 2 **Total Marks: 100**

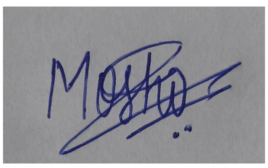
ANNEXURE A

LABORATORY PERFORMANCE QUESTIONS

Code of Conduct:

I Moshika M - 192424064 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Wt. %	**Level 4 Exemplary** 100% of weightage	**Level 3 Proficient** 75% of weightage	**Level 2 Developing** 50% of weightage	**Level 1 Beginning** 25% of weightage
Problem Understanding	**20%**	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	**25%**	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	**20%**	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	**20%**	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	**15%**	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases
Total	**100%**				

JAVA PROGRAMMING
LABORATORY PERFORMANCE QUESTIONS

1. Develop a simple Applet that displays your name, roll number, and course in different colors and fonts. Use the paint() method for rendering.

2. Create two types of Applets: a) Stand-alone Applet b) HTML embedded Applet. Demonstrate both and explain the differences.

3. Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.

4. Develop a Swing application that demonstrates the use of the following layout managers: FlowLayout, BorderLayout, and GridLayout. Switch among layouts using radio buttons.

5. Create a GUI using AWT controls (Button, Label, TextField, Checkbox, Choice). Perform simple operations such as registration form input and display the entered data.

6. Develop a program that demonstrates event handling using the Delegation Event Model. Handle at least two types of events (ActionEvent, ItemEvent) using listener interfaces.

7. Write a program that demonstrates different sources of events (Button click, TextField input, Checkbox selection, Window closing). Display the event source and type.

8. Create a menu-driven Swing application for a simple calculator (Add, Subtract, Multiply, Divide, Clear, Exit) using MenuBar, Menu, and MenuItem.

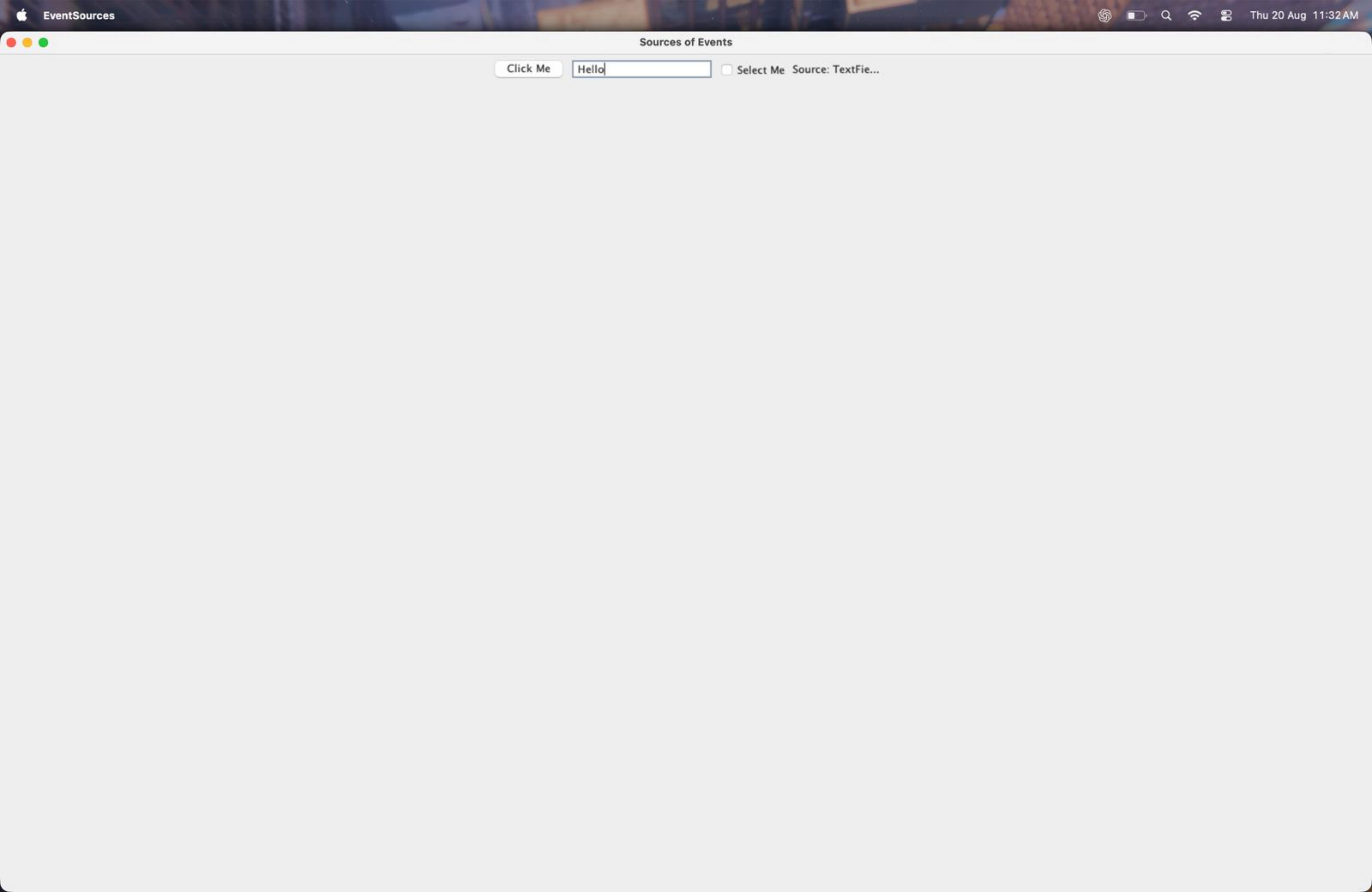
9. Develop an Applet that draws basic shapes (line, rectangle, oval) and changes their color based on user selection from a drop-down list (Choice control).

10. Design a GUI application that allows the user to maintain a simple student record (add /update/delete/display) using appropriate layout manager, controls, and event handling.

Q1 **sources of events** 25 marks

Write a program that demonstrates different sources of events (Button click, TextField input, Checkbox selection, Window closing). Display the event source and type

```java
import java.awt.*;
import java.awt.event.*;

public class EventSources extends Frame
        implements ActionListener, ItemListener, TextListener {

    Button button;
    TextField textField;
    Checkbox checkbox;
    Label result;

    public EventSources() {
        setTitle("Sources of Events");
        setSize(450, 250);
        setLayout(new FlowLayout());

        button = new Button("Click Me");
        textField = new TextField(20);
        checkbox = new Checkbox("Select Me");
        result = new Label("Perform an action");

        add(button);
        add(textField);
        add(checkbox);
        add(result);

        button.addActionListener(this);
        textField.addTextListener(this);
        checkbox.addItemListener(this);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                result.setText("Source: Window | Type: WindowClosing");
                dispose();
            }
        });

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        result.setText("Source: Button | Type: ActionEvent");
    }

    public void textValueChanged(TextEvent e) {
        result.setText("Source: TextField | Type: TextEvent");
    }

    public void itemStateChanged(ItemEvent e) {
        result.setText("Source: Checkbox | Type: ItemEvent");
    }

```

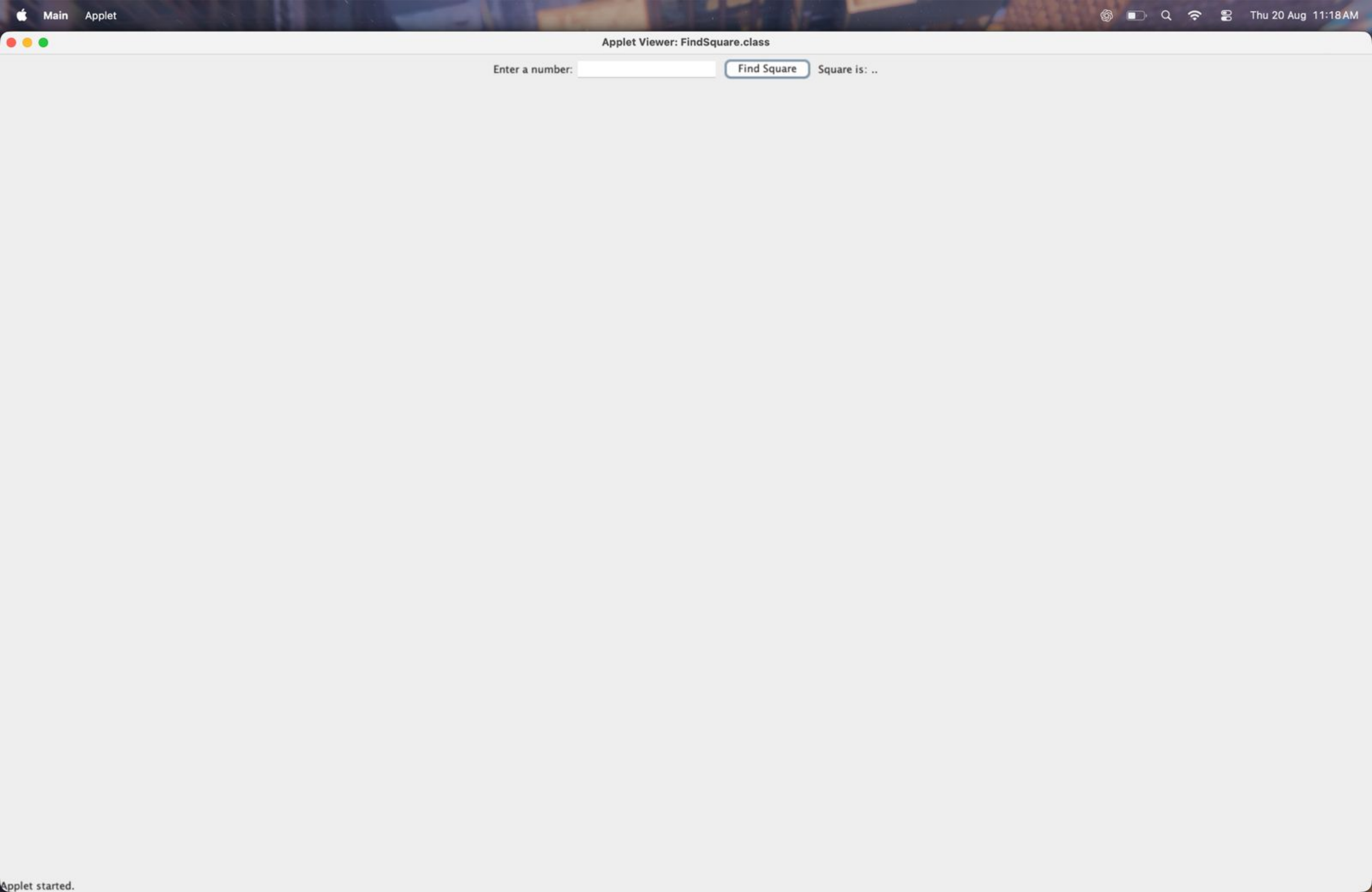

EventSources
Thu 20 Aug 11:32 AM
Sources of Events
Click Me
Hello
Select Me
Source: TextFie...

Q2 Find square

25 marks

Design an Applet that accepts a number from the user and displays its square when a button is clicked. Use AWT controls and appropriate event handling.

```java
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class FindSquare extends Applet implements ActionListener {

    Label l1, l2, l3;
    TextField tf;
    Button b;

    public void init() {
        setLayout(new FlowLayout());

        l1 = new Label("Enter a number:");
        tf = new TextField(20);
        b = new Button("Find Square");
        l2 = new Label("Square is:");
        l3 = new Label("");

        add(l1);
        add(tf);
        add(b);
        add(l2);
        add(l3);

        b.addActionListener(this);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            int num = Integer.parseInt(tf.getText());
            int square = num * num;
            l3.setText("" + square);
        } catch (NumberFormatException ex) {
            l3.setText("Invalid Number");
        }
    }
}
```

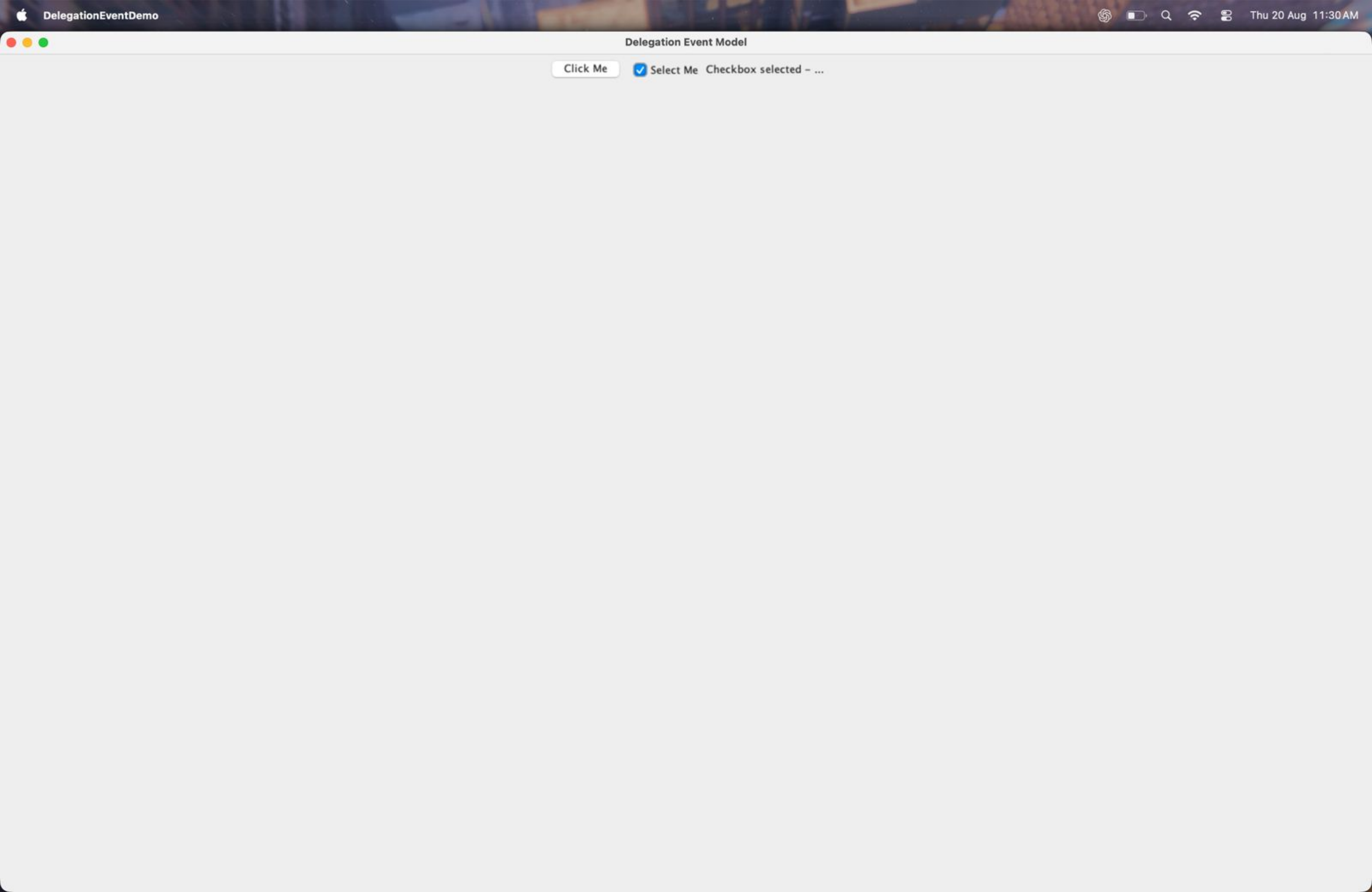

Main
Applet
Thu 20 Aug 11:18 AM
Applet Viewer: FindSquare.class
Enter a number:
Find Square
Square is: ..
Applet started.

Q3 Delegation Event Model

25 marks

Develop a program that demonstrates event handling using the Delegation Event Model. Handle at least two types of events (ActionEvent, ItemEvent) using listener interfaces.

```java
import java.awt.*;
import java.awt.event.*;

public class DelegationEventDemo extends Frame
        implements ActionListener, ItemListener {

    Button button;
    Checkbox checkbox;
    Label result;

    public DelegationEventDemo() {
        setTitle("Delegation Event Model");
        setSize(400, 250);
        setLayout(new FlowLayout());

        button = new Button("Click Me");
        checkbox = new Checkbox("Select Me");
        result = new Label("Select or click a control");

        add(button);
        add(checkbox);
        add(result);

        button.addActionListener(this);
        checkbox.addItemListener(this);

        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });
    }

    public void actionPerformed(ActionEvent e) {
        result.setText("Button clicked - ActionEvent handled");
    }

    public void itemStateChanged(ItemEvent e) {
        if (checkbox.getState())
            result.setText("Checkbox selected - ItemEvent handled");
        else
            result.setText("Checkbox deselected - ItemEvent handled");
    }

    public static void main(String[] args) {
        new DelegationEventDemo();
    }
}
```

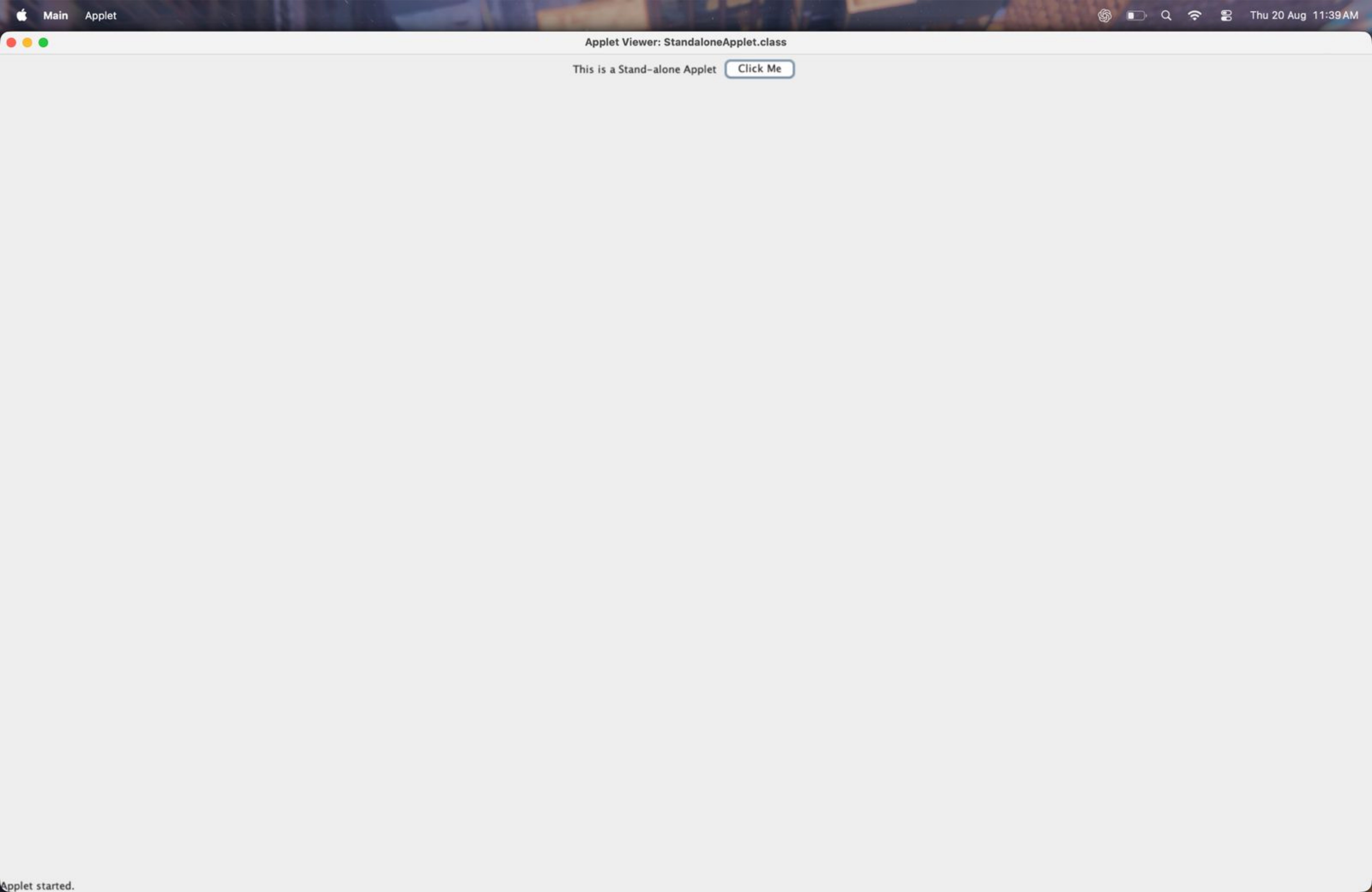

DelegationEventDemo
Thu 20 Aug 11:30 AM
Delegation Event Model
Click Me
Select Me
Checkbox selected - ...

Q4 two types of Applets

25 marks

Create two types of Applets: a) Stand-alone Applet b) HTML embedded Applet. Demonstrate both and explain the differences.

```java
import java.applet.Applet;
import java.awt.*;

public class StandaloneApplet extends Applet {

    public void init() {
        setLayout(new FlowLayout());
        add(new Label("This is a Stand-alone Applet"));
        add(new Button("Click Me"));
    }
}ja
```


Main
Applet
Thu 20 Aug 11:39 AM
Applet Viewer: StandaloneApplet.class
This is a Stand-alone Applet
Click Me
Applet started.

```java
import java.applet.Applet;
import java.awt.*;

public class HTMLEmbeddedApplet extends Applet {

    public void init() {
        setLayout(new FlowLayout());
        add(new Label("This Applet is Embedded in HTML"));
        add(new Button("Submit"));
    }
}
```

Main Applet

Thu 20 Aug 11:41 AM

Applet Viewer: HTMLEmbeddedApplet.class

This Applet is Embedded in HTML Submit

Applet started.